Research Project COCONUT: CO-FOOTBALL

Paper type: System Description Paper

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Abstract

Modern football has become increasingly tactically complex, making it difficult for non-expert fans to understand and reason about strategic decisions made by elite teams. This paper presents *CO-FOOTBALL*, a co-creative system designed to support football fans in exploring contemporary football tactics through human–AI collaboration. The system combines data-driven mismatch analysis based on player attributes with interactive visual representations of team formations and a mixed-initiative conversational AI assistant that provides tactical explanations and suggestions. Rather than acting as an autonomous tactic generator, *CO-FOOTBALL* guides users through a realistic tactical exploration process while preserving user agency. We report a preliminary qualitative user study with football fans, which suggests that the system is perceived as both useful and co-creative, supporting tactical understanding and exploratory decision-making. These results indicate that mixed-initiative AI systems can effectively support engagement with complex tactical domains such as modern football.

1 Introduction

Motivation

Football is an inherently complex sport, and this complexity has become even more pronounced in recent years. The tactical dimension of the game has gained increasing importance and, in many situations, can be more decisive than the individual technical quality of the players involved. At the highest level, football tactics have grown substantially more sophisticated, and individual technical ability alone is no longer sufficient to guarantee success. Top teams are well aware of this evolution, which is why the presence of an elite manager has become a critical factor in modern football.

The tactical aspect of football is particularly complex because effective strategies cannot be designed in isolation. Instead, a tactical plan must be tailored to the specific characteristics of a team, including the players available, their playing styles, and their physical attributes, as well as the strengths and weaknesses of the opposing team. A tactical plan is only effective if it is feasible: it must be executable by the players on the pitch and appropriate for exploiting the opponent's vulnerabilities while limiting their strengths.

In recent years, elite teams have gained access to advanced technologies that optimize training and maintain players in peak physical condition. As a result, physical attributes such as strength, stamina, height, and speed play an increasingly decisive role in match outcomes, making the identification of physical mismatches a crucial component of modern tactical planning.

Research Goal

CO-FOOTBALL is a co-creative system designed to help football fans explore and better understand contemporary tactical decision-making through human–AI collaboration (Davis et al. 2015). As modern football has become increasingly complex, many spectators lack the expertise needed to interpret strategic choices made by elite teams.

The system aims to enhance tactical understanding by enabling users to explore ideas, develop alternative strategies, and validate hypotheses they are uncertain about. Rather than acting as a simple tactic generator, *CO-FOOTBALL* guides users through a realistic tactical exploration process, keeping suggestions feasible and grounded in real-world constraints while preserving the user's intended playing style and overall direction.

Contributions

This paper introduces *CO-FOOTBALL*, a co-creative system that supports football fans in exploring contemporary football tactics through human–AI collaboration. The main contributions are as follows:

- An interactive web-based system for tactical exploration that allows users to configure teams, formations, and lineups in an iterative manner.
- An automated method for detecting player-level and positional mismatches based on technical and physical attributes, providing a data-driven basis for tactical reasoning.
- A mixed-initiative conversational AI coach that engages users in dialogue to generate, explain, and adapt tactical ideas while preserving user agency.
- Visual representations of team formations that support spatial reasoning about tactical structures and immediate feedback on user decisions.

- A preliminary user study with football fans evaluating usability, usefulness, and perceived co-creativity.

2 Method

System Overview

CO-FOOTBALL is implemented as a modular system composed of a web-based frontend, a backend for data processing and tactical analysis, and an AI component that supports conversational interaction. This architecture is designed to enable iterative, mixed-initiative co-creation between the user and the system.

The frontend is developed using *Streamlit* and serves as the primary interface through which users interact with the system. Through this interface, users can select teams and formations, perform substitutions, modify line-ups, and input natural language queries to communicate with the AI coach. The frontend also provides visual feedback by displaying graphical representations of team formations, allowing users to observe how tactical changes affect the spatial organization of players on the pitch.

The backend is responsible for all core computational logic. It includes functions for filtering and processing player data, generating team line-ups based on selected formations, computing player-level and positional mismatches between opposing teams, and producing visual plots of formations that reflect team structure, player positions, and visual attributes such as team colors. These components ensure that tactical analyses remain grounded in the selected teams and configurations.

A central component of the architecture is the conversational AI module, which enables tactical reasoning and explanation. The system integrates a large language model (LLM) accessed via the Groq API, which functions as the AI coach (Brown and others 2020). This module receives as input the detected mismatches, the user-defined line-ups, and a pre-defined prompt that encodes tactical reasoning guidelines. Based on these inputs, the model generates an initial tactical proposal. Users can then iteratively interact with the AI coach to ask questions, request clarifications, or steer the tactical direction, with the system adapting its responses accordingly. This interaction loop supports a co-creative process in which human insight and AI-generated analysis jointly contribute to tactical exploration.

Data Representation

One of the main challenges in developing *CO-FOOTBALL* was the identification of suitable player data, which are not always readily available in a structured and comprehensive form. Moreover, the system requires quantitative representations of attributes that are sometimes qualitative in nature, such as aerial ability or balance. To address these challenges, we adopted player statistics from *EA Sports FC 26*, extracted from the player database published on Kaggle (Flynn28 2025). These statistics are designed to approximate real-world player capabilities and, while not perfectly accurate, they provide a widely used and reasonable proxy for player characteristics in an initial system implementation.

The selected dataset provides quantitative representations of both physical and technical attributes, enabling data-driven tactical analysis. In addition, it includes information about the positions a player has covered throughout their career, as well as secondary positions the player is likely capable of performing beyond their primary role. This positional modeling is crucial for evaluating the feasibility of tactical adaptations, as it allows the system to assess whether redeploying a player in an alternative position is realistic. Consequently, this information directly informs the system's decision on whether to suggest positional changes during the co-creative tactical exploration process.

Mismatch Analysis

CO-FOOTBALL identifies tactical advantages and vulnerabilities through an automated mismatch analysis that operates at both the individual and positional level. The goal of this component is to extract interpretable, data-driven insights that can support tactical reasoning and guide the subsequent co-creative interaction between the user and the system.

The analysis begins by abstracting individual player roles into higher-level positional zones. This mapping allows the system to normalize different formation-specific role labels and enables consistent comparisons across tactical setups. Based on these zones, the system defines a set of predefined match-up rules that specify which opposing roles are likely to interact directly during a match. For example, wide attacking players are compared against corresponding defensive roles, center forwards are matched with central defenders, and attacking midfielders are compared with central midfielders.

For each valid attacker–defender pairing, the system evaluates a set of composite performance metrics derived from the underlying player attributes. These metrics capture key dimensions of modern football performance, including speed, aerial ability, physicality, and balance. Each metric is computed by aggregating relevant low-level attributes (e.g., sprint speed and acceleration for speed, height and jumping for aerial ability), providing an interpretable representation of higher-level footballing qualities.

A mismatch is detected when the difference between two opposing players on a given metric exceeds a predefined threshold, indicating a potentially meaningful advantage or weakness. The analysis is performed symmetrically for both teams, ensuring that advantages for one side are also reflected as vulnerabilities for the other. The resulting set of mismatches is stored in a structured format and forms the quantitative basis for the reasoning process of the conversational AI coach.

AI Tactical Assistant

CO-FOOTBALL integrates a large language model (LLM) as an AI Tactical Assistant to support tactical interpretation, explanation, and iterative refinement through natural language interaction. Rather than autonomously generating tactics, the assistant functions as a mixed-initiative collaborator that reasons over structured inputs and responds to user-driven exploration.

The assistant operates by interpreting the structured output of the mismatch analysis module, which encodes key individual and positional advantages between the two teams. These mismatches, together with the selected formations, starting line-ups, and available bench players, are combined into a textual match context that constrains the model's reasoning space. This context integrates quantitative information (e.g., metric differences and player roles) with qualitative tactical instructions, ensuring that generated suggestions remain grounded in realistic football principles.

Based on this constrained context, the LLM produces an initial tactical proposal covering attacking strategies, defensive organization, team principles, and potential substitutions. The prompt design enforces domain-specific constraints, such as limiting substitutions, preventing the invention of players, and explicitly linking recommendations to detected mismatches, thereby prioritizing interpretability over unconstrained text generation.

The AI Tactical Assistant further supports a conversational interaction loop that enables iterative co-creation. Users may ask follow-up questions, challenge assumptions, or redirect the tactical approach, with each intervention incorporated into subsequent responses. This dialogue-based process positions the assistant as a responsive collaborator rather than an authoritative decision-maker, supporting realistic, data-informed tactical exploration.

User Interaction Design

CO-FOOTBALL is implemented with a graphical user interface designed to support intuitive and progressive user interaction throughout the tactical exploration process. The interaction flow is structured to guide users from high-level configuration choices to more detailed tactical reasoning, while maintaining transparency and user agency.

The interaction begins with team selection. Users can choose the team they wish to control from clubs belonging to the top seven European leagues (Spain, France, Portugal, the Netherlands, England, Italy, and Germany), and select a formation from a predefined set (e.g., 4–3–3, 3–5–2). The same configuration process is then repeated for the opposing team, allowing users to define both sides of the match context.

Once both teams and formations are selected, the system generates graphical plots of the two starting line-ups. These visualizations depict player positions on a realistic pitch layout, enabling users to immediately understand how each team is structured spatially. At this stage, users can evaluate the default line-ups generated by the system and decide whether to make tactical adjustments.

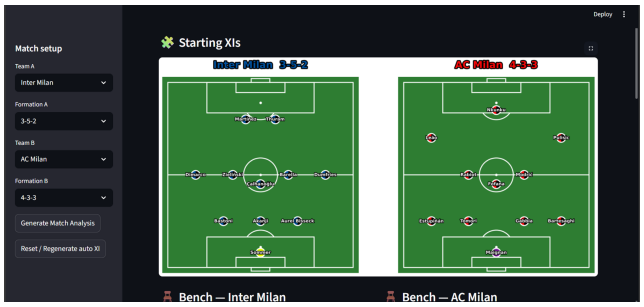

Figure 1: Graphical user interface of *CO-FOOTBALL* showing team selection, formation configuration, and the visual representation of the generated starting line-ups.

Users may then perform substitutions or positional changes within the starting eleven, selecting players from the available bench options. Each modification is immediately reflected in the visual representation, reinforcing the connection between user actions and tactical consequences. This interactive manipulation of line-ups allows users to experiment with alternative configurations before engaging in deeper tactical reasoning.

Finally, users can initiate a dialogue with the AI Tactical Assistant to request tactical plans, ask questions, or propose alternative ideas. This conversational interaction builds upon the previously defined teams, formations, and line-ups, ensuring that the AI's responses remain consistent with the user's choices.

3 Results

User Study

To evaluate the effectiveness of *CO-FOOTBALL* and its support for co-creativity, we conducted a preliminary qualitative user study involving four football fans. Participants were first asked to interact with the system by configuring teams and formations, performing substitutions, and engaging in dialogue with the AI Tactical Assistant. After this hands-on interaction, they completed a structured questionnaire and were optionally invited to provide open-ended feedback.

The questionnaire consisted of seven Likert-scale items designed to assess perceived co-creativity, user control, interaction quality, and overall usefulness of the system. Participants indicated their level of agreement with each statement on a five-point scale, where 1 corresponded to *strongly disagree* and 5 to *strongly agree*. The questionnaire was organized into three sections: *Perceived Co-Creativity* (three items), *Control & Interaction* (two items), and *Usefulness & Engagement* (two items). One interaction-timing question was excluded from the analysis to better align the evaluation with the study's focus on co-creativity and user agency.

Table 1 summarizes the responses for each participant. Overall, results show consistently positive evaluations across all dimensions. Participants generally perceived the system as a creative collaborator that influenced their tactical thinking, while still allowing them to retain control over the overall tactical direction. High agreement scores were also observed for items related to usefulness and engagement.

Table 1: User study questionnaire results (Likert scale 1–5), including mean scores.

Question	P1	P2	P3	P4	Mean
Perceived Co-Creativity					
Q1. Creative collaborator	4	4	5	4	4.25
Q2. Influenced my decisions	3	4	5	5	4.25
Q3. Proposed novel strategies	5	3	5	5	4.50
Control & Interaction					
Q4. Felt in control	5	4	5	4	4.50
Q5. Interaction felt natural	4	5	4	4	4.25
Usefulness & Engagement					
Q6. Helped explore tactics	4	3	5	5	4.25
Q7. Would use again	4	5	4	4	4.25

In addition to the scaled questions, participants answered two optional open-ended questions. When asked what they found most useful or interesting about the system, respondents highlighted its ability to validate their own tactical intuitions, such as assessing whether a potential substitution or tactical adjustment would be reasonable during a live match. Others emphasized the usefulness of the system for preparing tactical plans against unfamiliar opponents, where limited prior knowledge would otherwise hinder informed decision-making.

Regarding potential improvements, participants suggested incorporating more advanced or dynamic statistics, such as players' current form, to increase realism. Another recurring suggestion was to provide initial recommendations for starting line-ups and formations, particularly for opposing teams that users may not be familiar with. These insights suggest promising directions for future development.

Although limited in scale, this preliminary user study provides initial evidence that *CO-FOOTBALL* is perceived as both useful and co-creative, and it offers qualitative insights that can inform future iterations and more extensive evaluations.

4 Discussion

Interpretation of Results

The results indicate that, as a first implementation, *CO-FOOTBALL* functions effectively as a co-creative system within its intended usage domain and target audience. Participants consistently reported positive perceptions of the system's ability to act as a creative collaborator, influence tactical thinking, and support exploratory decision-making while preserving user control. These findings suggest that the system successfully aligns with its primary research goal of fostering human–AI co-creativity in football tactical analysis.

In particular, the high agreement scores related to usefulness and engagement indicate that AI-generated tactical explanations can meaningfully support users in understanding complex tactical concepts. This suggests that mixed-initiative AI systems can act as effective educational and exploratory tools by presenting data-driven insights in a clear and more accessible manner.

Limitations

Despite these encouraging results, several limitations must be acknowledged. A primary limitation concerns the data source used by the system. Player attributes are derived from a commercial video game dataset, which, while offering a structured and comprehensive approximation of real-world player characteristics, does not fully capture the complexity and variability of professional football data. Access to real-world datasets collected by professional clubs (potentially including medical information, workload measures, and indicators of current form) would significantly improve the realism and accuracy of the system's analyses.

Additionally, the user study was limited in scale, involving a small number of participants and focusing on qualitative and descriptive measures. As such, the findings should be interpreted as preliminary.

5 Conclusion

This paper presented *CO-FOOTBALL*, a co-creative system aimed at supporting football fans in exploring modern football tactics through data-driven analysis and mixed-initiative AI interaction. By integrating automatic mismatch detection, interactive visualizations, and conversational tactical explanations, the system enables accessible yet realistic tactical reasoning while preserving user agency. A preliminary user study suggests that *CO-FOOTBALL* is perceived as both useful and co-creative, supporting exploratory decision-making in a complex and dynamic domain.

Future Work

Several directions could be explored to further enhance the system. One potential improvement concerns the integration of richer and more dynamic datasets, such as real-world performance data or indicators of player form, which could increase the realism and analytical depth of the tactical analysis. The mismatch detection mechanism could also be extended to incorporate additional contextual features, enabling more nuanced comparisons between players. From an evaluation perspective, future studies could involve larger and more diverse user samples to better assess co-creativity, usability, and learning outcomes.

References

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